

Cricket Knowledge Base

12-Page Test Document for Retrieval-Augmented Generation (RAG)

This document covers cricket formats, batting, bowling, fielding, rules, strategy, statistics, analytics, and RAG test questions. It is designed to provide factual, numerical, semantic, and reasoning queries for a cricket-focused RAG application.

Recommended tests: direct retrieval, paraphrased questions, numerical reasoning, comparisons, and multi-section questions.

1. Introduction to Cricket

Overview

Cricket is a bat-and-ball team sport played between two teams of eleven players. The objective is to score more runs than the opposing team while restricting the opponent's scoring opportunities. Cricket has a common foundation across formats, but match duration, tactics, and tempo vary considerably.

Playing Area

A cricket ground normally contains an oval outfield with a rectangular central pitch. At each end is a wicket consisting of three stumps and two bails. The striker faces the bowler while the non-striker stands at the opposite end.

Scoring

Runs can be scored by running between wickets, hitting boundaries, or through extras. A ball reaching the boundary after bouncing normally scores four runs, while a ball cleared directly over the boundary normally scores six.

RAG Test Value

Cricket is useful for RAG testing because it contains related entities, numbers, rules, statistics, and tactical concepts. A good system should distinguish batting, bowling, rules, formats, and statistics.

2. Cricket Formats

Test Cricket

Test cricket is the longest major international format. A Test is scheduled across up to five days and gives each side the opportunity to bat twice, subject to the match situation. Teams balance scoring, wickets, and time.

One Day International Cricket

ODI cricket uses a limited-overs structure. Each team normally has one innings and faces a fixed number of overs. Wickets, run rate, fielding restrictions, and overs remaining all influence strategy.

T20 Cricket

Twenty20 cricket is a short, high-intensity format. Each team normally faces a maximum of 20 overs. Batters often prioritize boundary scoring and rapid strike rotation.

Format Comparison

Tests emphasize endurance and multi-session strategy, ODIs combine longer planning with a fixed overs limit, and T20s emphasize rapid decisions and aggressive scoring.

3. Batting and Run Scoring

Batting Fundamentals

A batter protects the wicket, scores runs, and responds to bowling tactics. Shot selection depends on line, length, pace, movement, field placement, and match situation.

Strike Rate

Strike rate is commonly calculated as runs divided by balls faced, multiplied by 100. A batter scoring 80 runs from 50 balls has a strike rate of 160.00.

Boundaries

Fours and sixes can change the momentum of an innings. Statistical definitions should always be retrieved from the relevant source because analysts may define boundary-related percentages differently.

Partnerships

Partnerships measure the combined contribution of two batters. They can stabilize an innings after wickets, accelerate scoring, or provide a strong finish.

4. Bowling

Bowling Objective

A bowler attempts to dismiss batters and restrict scoring. Bowlers vary line, length, pace, swing, seam movement, spin, trajectory, and release point.

Pace Bowling

Fast and medium-fast bowlers can create difficulty through speed and movement. Swing occurs through the air, while seam movement can occur after pitching. Yorkers target the area near the batter's feet.

Spin Bowling

Spin bowlers use changes in rotation and trajectory. Off-spin, leg-spin, left-arm orthodox spin, and wrist spin create different angles and movement patterns.

Economy Rate

Economy rate measures runs conceded per over. A bowler conceding 30 runs in 5 overs has an economy rate of 6.00.

5. Fielding and Wicketkeeping

Fielding Roles

Fielders combine prevention, catching, throwing, and tactical positioning. Close fielders create catching opportunities while boundary fielders try to stop fours and sixes.

Catching

A catch dismisses a batter when a fielder legally completes the catch before the ball becomes dead, subject to the Laws of Cricket.

Wicketkeeping

The wicketkeeper stands behind the striker's wicket and collects deliveries, attempts stumpings, supports catches, and communicates with the bowler and captain.

Run Outs

A run-out can occur when a batter is outside the relevant crease and the wicket is fairly put down. Accurate throws can create quick dismissal opportunities.

6. Cricket Rules and Dismissals

Common Dismissals

Common dismissals include bowled, caught, leg before wicket, run out, stumped, hit wicket, and obstructing the field.

Bowled

A batter is bowled when a legal delivery breaks the wicket at the striker's end, subject to the relevant Laws.

Caught

A caught dismissal occurs when a fielder catches the ball before it touches the ground, subject to the requirements of the Laws.

LBW

Leg before wicket depends on conditions including where the ball pitched, where it struck the batter, and whether it would have hit the wicket.

7. Extras and Match Scoring

Types of Extras

Common extras include wides, no-balls, byes, and leg byes. Extras contribute to the team total but are generally not credited to the batter's individual score.

Wides

A wide is generally called when a delivery is outside the batter's reach according to applicable playing conditions. A wide does not count as a legal ball in an over.

No-Balls

A no-ball is called when the bowler breaches a relevant delivery condition. In many limited-overs competitions, a no-ball can also create a free-hit opportunity.

Byes and Leg Byes

Byes are runs completed when the ball passes the batter without being hit by the bat in the relevant circumstances. Leg byes are scored when the ball touches the batter's person under permitted conditions.

8. Match Strategy

Powerplay Strategy

Limited-overs cricket can include fielding restrictions during designated periods. Batting teams may attack during these phases while bowlers attempt to exploit the increased risk of aggressive batting.

Middle Overs

The middle phase often involves balancing boundary attempts with strike rotation. Spinners may control scoring while batters look for gaps and favorable matchups.

Death Overs

Final overs are often high-scoring. Batters target boundaries while bowlers use yorkers, slower balls, wide lines, and field placements to reduce clean contact.

Chasing a Target

A chasing team manages required run rate and wickets in hand. A target can be pursued through partnerships rather than continuous boundary hitting.

9. Cricket Statistics

Batting Statistics

Important batting statistics include runs, average, strike rate, boundaries, balls faced, and dismissals. No single statistic fully describes performance.

Bowling Statistics

Bowling analysis includes wickets, runs conceded, overs, economy rate, average, strike rate, maidens, and dot-ball percentage.

Team Statistics

Team analysis can include total runs, wickets lost, run rate, required run rate, extras, boundary count, and partnerships.

Example

A batter scoring 96 from 72 balls has a strike rate of 133.33. Another scoring 70 from 40 balls has a higher strike rate, but match context determines which innings was more valuable.

10. Cricket Tactics and Matchups

Left-Hand and Right-Hand Combinations

Teams may use different batting handedness because bowling angles and field positions change. Captains can also change bowling plans to create favorable matchups.

Spin versus Pace

The choice between spin and pace depends on pitch behavior, batter tendencies, match phase, and conditions.

Aggression versus Control

Aggressive batting increases boundary opportunities but can increase dismissal risk. Defensive batting preserves wickets but may allow the required rate to rise.

Decision Making

Strong tactical decisions combine recent scoring patterns, matchups, field positions, pitch conditions, and historical performance.

11. Cricket Analytics and RAG

Why Cricket Works for RAG

Cricket documents contain facts distributed across rules, statistics, formats, tactics, and player analysis. RAG can retrieve relevant passages before an LLM generates an answer.

Chunking

A cricket knowledge base can be divided into topics such as batting, bowling, rules, fielding, and statistics. Chunk size and overlap affect retrieval quality.

Metadata

Useful metadata includes topic, format, year, competition, and source. Metadata filtering can improve targeted retrieval.

RAG Test Design

A strong test set should include direct questions, paraphrased questions, numerical questions, comparison questions, and multi-hop questions.

12. RAG Test Questions

Direct Questions

What is the maximum number of overs in a normal T20 innings? How is strike rate calculated? What does economy rate measure? Name four common extras. What is the role of the wicketkeeper?

Reasoning Questions

Why might a chasing team avoid excessive risk early? Why should strike rate be interpreted in context? How can fielding restrictions influence batting? Why should economy be considered alongside wickets?

Numerical Questions

A batter scores 75 from 50 balls. What is the strike rate? A bowler concedes 28 runs in 4 overs. What is the economy rate? A team scores 180 in 20 overs. What is its run rate?

Final Challenge

Compare Test and T20 cricket in terms of duration, innings structure, scoring approach, and strategic priorities. Explain why strike rate should not be evaluated without match context.